Chapter 7. Regression Estimation

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1 The Regression Estimator

The ratio estimator works best when y is proportional to x (a line through the origin). If the relation is linear with a nonzero intercept, use the regression line instead. As in Ch. 6 the x's are treated as fixed (already observed) and the y's as random:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \beta_0' + \beta_1(x_i - \bar{x}) + \varepsilon_i, \qquad \hat{\beta}_0' = \bar{y}, \quad \hat{\beta}_1 = b = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2},$$

so the fitted mean response is $\widehat{\mathrm{E}}(y \mid x) = \bar{y} + b(x - \bar{x})$. Evaluating it at the known population mean μ_x gives the estimator.

Definition 7.1

The **regression estimator** of μ_y is

$$\hat{\mu}_{yL} = \widehat{\mathrm{E}}(y \mid \mu_x) = \bar{y} + b(\mu_x - \bar{x}) = \frac{1}{n}\sum_{i=1}^{n} [y_i - b(x_i - \mu_x)],$$

i.e. the sample mean of the adjusted values $y_i - b(x_i - \mu_x)$.

If the sample happens to have $\bar{x} < \mu_x$ and $b > 0$, the sample mean $\bar{y}$ is probably too small as well, and $\hat{\mu}_{yL}$ corrects it upward by $b(\mu_x - \bar{x})$.

Theorem 7.2: Model unbiasedness

Under the model $\mathrm{E}(y_i) = \beta_0 + \beta_1 x_i$ with fixed x's,

$$\mathrm{E}(\hat{\mu}_{yL}) = \beta_0 + \beta_1 \mu_x = \mathrm{E}(y \mid \mu_x).$$

Proof. $\mathrm{E}(\bar{y}) = \beta_0 + \beta_1 \bar{x}$ and $\mathrm{E}(b) = \beta_1$ (Ch. 6), so $\mathrm{E}(\hat{\mu}_{yL}) = \beta_0 + \beta_1 \bar{x} + \beta_1(\mu_x - \bar{x}) = \beta_0 + \beta_1 \mu_x$. ■

2 Variance of the Regression Estimator

2.1 Fixed slope

First take a fixed constant β and $\bar{y}_{\text{reg}} = \bar{y} + \beta(\mu_x - \bar{x})$. This is the SRS mean of $u_i = y_i - \beta(x_i - \mu_x)$, whose population mean is μ_y, so it is exactly unbiased.

Theorem 7.3

$$\operatorname{Var}(\bar{y}_{\text{reg}}) = \frac{1-f}{n}(S_y^2 - 2\beta S_{xy} + \beta^2 S_x^2),$$

which is minimized at $\beta = S_{xy}/S_x^2$ (the population LS slope), where

$$\operatorname{Var}(\bar{y}_{\text{reg}}) = \frac{1-f}{n}\Big(S_y^2 - \frac{S_{xy}^2}{S_x^2}\Big) = \frac{1-f}{n} S_y^2(1-\rho^2).$$

Proof. By Ch. 2, $\operatorname{Var}(\bar{y}_{\text{reg}}) = \frac{1-f}{n} S_u^2$ with

$$\begin{aligned} S_u^2 &= \frac{1}{N-1}\sum_{i=1}^{N}\big[(y_i - \mu_y) - \beta(x_i - \mu_x)\big]^2 \\ &= \frac{1}{N-1}\Big[\sum (y_i-\mu_y)^2 - 2\beta\sum (x_i-\mu_x)(y_i-\mu_y) + \beta^2 \sum (x_i-\mu_x)^2\Big] = S_y^2 - 2\beta S_{xy} + \beta^2 S_x^2. \end{aligned}$$

This quadratic in β is minimized at $\beta = S_{xy}/S_x^2$, giving $S_y^2 - S_{xy}^2/S_x^2 = S_y^2(1-\rho^2)$. ∎

2.2 Estimated slope

With b estimated, write $\xi_i = (y_i - \mu_y) - \beta(x_i - \mu_x)$ for $\beta = S_{xy}/S_x^2$, so $y_i = \mu_y + \beta(x_i - \mu_x) + \xi_i$. Then

$$b = \frac{\sum (x_i - \bar{x}) y_i}{\sum (x_i - \bar{x})^2} = \beta \frac{\sum (x_i - \bar{x})(x_i - \mu_x)}{\sum (x_i - \bar{x})^2} + \frac{\sum (x_i - \bar{x})\xi_i}{\sum (x_i - \bar{x})^2} = \beta + \frac{\sum (x_i - \bar{x})\xi_i}{\sum (x_i - \bar{x})^2},$$

and the last term is of smaller order since the ξ_i are uncorrelated with x in the population. Hence $b \approx \beta$ in large samples and

$$\operatorname{Var}(\hat{\mu}_{yL}) \approx \operatorname{Var}(\bar{y}_{\text{reg}}) = \frac{1-f}{n} S^2, \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}\big[(y_i - \mu_y) - \beta(x_i - \mu_x)\big]^2.$$

Theorem 7.4: Estimated variance

$$\widehat{\operatorname{Var}}(\hat{\mu}_{yL}) = \frac{N-n}{Nn} MS_E, \qquad MS_E = \frac{1}{n-2}\sum_{i=1}^{n}\big[(y_i - \bar{y}) - b(x_i - \bar{x})\big]^2,$$

the mean square error from the simple linear regression of y on x.

The sample analogue of S^2 is $\frac{1}{n-1}\sum e_i^2$ with residuals $e_i = (y_i - \bar{y}) - b(x_i - \bar{x})$; dividing by $n-2$ instead (two estimated coefficients) gives MS_E, and the difference is negligible for moderate n.

Example 7.5: Calculus marks

A mathematics achievement test was taken by all $N = 486$ entering students, with mean $\mu_x = 52$. For $n = 10$ sampled students:

Student	1	2	3	4	5	6	7	8	9	10
Test (x)	39	43	21	64	57	47	28	75	34	52
Final (y)	65	78	52	82	92	89	73	98	56	75

Here $\bar{x} = 46$, $\bar{y} = 76$, and the R fit gives $\hat{y} = 40.78 + 0.7656x$ with residual standard error 8.704 on 8 df ($MS_E = 75.76$, $R^2 = 0.705$). Then

$$\hat{\mu}_{yL} = 76 + 0.7656(52 - 46) = 80.6,$$

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yL}) = \frac{486 - 10}{486 \times 10}(75.76) = 7.42, \qquad B = 2\sqrt{7.42} = 5.45.$$

The fitted intercept (40.8) is far from zero, so a ratio estimator would be inappropriate here: $r\mu_x = (760/460)(52) = 85.9$ overshoots.

3 Relative Efficiency: Mean, Ratio and Regression

For two estimators E_1, E_2 of μ_y, $\mathrm{RE}(E_1/E_2) = \mathrm{Var}(E_2)/\mathrm{Var}(E_1)$; if it exceeds 1, E_1 is preferred. The estimated version (with estimated variances) is meaningful for large samples. The three estimated variances share the factor $\frac{1-f}{n}$:

$$\widehat{\mathrm{Var}}(\bar{y}) = \frac{1-f}{n}s_y^2, \qquad \widehat{\mathrm{Var}}(\hat{\mu}_y) = \frac{1-f}{n}\left(s_y^2 + r^2 s_x^2 - 2r\hat{\rho}s_x s_y\right), \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{yL}) = \frac{1-f}{n}s_y^2(1-\hat{\rho}^2).$$

Derivation: the regression variance in correlation form

$$\begin{aligned}\frac{1}{n-1}\sum\left[(y_i - \bar{y}) - b(x_i - \bar{x})\right]^2 &= \frac{1}{n-1}\left[S_{yy} - 2b\,S_{xy} + b^2 S_{xx}\right] \\ &= s_y^2 - 2b\,s_{xy} + b^2 s_x^2 = s_y^2 - b^2 s_x^2,\end{aligned}$$

since $s_{xy} = b s_x^2$. Moreover $\hat{\rho} = \frac{s_{xy}}{s_x s_y} = \frac{b s_x^2}{s_x s_y}$ gives $\hat{\rho} s_y = b s_x$, so $s_y^2 - b^2 s_x^2 = s_y^2(1-\hat{\rho}^2)$.

3.1 Mean versus ratio

From Ch. 5,

$$\widehat{\mathrm{RE}}(\hat{\mu}_y/\bar{y}) = \frac{s_y^2}{s_y^2 + r^2 s_x^2 - 2r\hat{\rho}s_x s_y},$$

and the ratio estimator is more efficient iff $\hat{\rho} > \frac{1}{2}\cdot\frac{s_x/\bar{x}}{s_y/\bar{y}}$. Recall that r is generally biased for R; the bias is negligible when the relation is a line through the origin.

3.2 Mean versus regression

$$\widehat{\mathrm{RE}}(\hat{\mu}_{yL}/\bar{y}) = \frac{s_y^2}{s_y^2(1-\hat{\rho}^2)} = \frac{1}{1-\hat{\rho}^2} > 1 \quad \text{unless } \hat{\rho} = 0.$$

The regression estimator is (asymptotically) never worse than $\bar{y}$.

3.3 Ratio versus regression

Theorem 7.6

$$\widehat{\mathrm{RE}}(\hat{\mu}_{yL}/\hat{\mu}_y) = \frac{s_y^2 + r^2 s_x^2 - 2r\hat{\rho} s_x s_y}{s_y^2(1-\hat{\rho}^2)} \geq 1,$$

with equality iff $r = b$.

Proof. The numerator minus the denominator is

$$r^2 s_x^2 - 2r\hat{\rho} s_x s_y + \hat{\rho}^2 s_y^2 = (r s_x - \hat{\rho} s_y)^2 = (r s_x - b s_x)^2 = s_x^2 (r-b)^2 \geq 0.$$

■

Both r and b are linear in the y's: $r = \sum y_i / \sum x_i = \sum d_i y_i$ with equal weights $d_i = 1/\sum x_j$, while $b = \sum w_i y_i$ with $w_i = (x_i - \bar{x})/\sum (x_j - \bar{x})^2$. They coincide only when the LS line passes through the origin, which is exactly when the ratio estimator is as good as the regression estimator.

4 Selected exercises

Motivation. The first two items carry regression estimation into stratified sampling, in parallel with the separate and combined ratio estimators of Ch. 5; the last two give the exact design bias of $\hat{\mu}_{yL}$ and its interpretation as a reweighting of the sample.

Exercise 7.1 (Separate regression estimator)

In stratified random sampling with strata $h = 1, \ldots, L$, weights w_h and known stratum means μ_{xh}, the separate regression estimator is

$$\hat{\mu}_{yLS} = \sum_{h=1}^{L} w_h \left[\bar{y}_h + b_h (\mu_{xh} - \bar{x}_h) \right],$$

with b_h the LS slope in stratum h. Find its approximate variance and an estimator of it.

Solution.

Step 1. (Independence.) The term in brackets, $\hat{\mu}_{yL,h}$, is the regression estimator computed from the SRS in stratum h alone, and the strata are sampled independently, so

$$\mathrm{Var}(\hat{\mu}_{yLS}) = \sum_h w_h^2 \mathrm{Var}(\hat{\mu}_{yL,h}).$$

Step 2. (Each stratum.) By the estimated-slope argument of this chapter applied in stratum h, $b_h \approx \beta_h = S_{xyh}/S_{xh}^2$ for large n_h, and the fixed-slope theorem gives

$$\mathrm{Var}(\hat{\mu}_{yL,h}) \approx \frac{1-f_h}{n_h} S_{yh}^2 (1-\rho_h^2), \qquad \rho_h = \frac{S_{xyh}}{S_{xh} S_{yh}}.$$

Step 3. Hence

$$\mathrm{Var}(\hat{\mu}_{yLS}) \approx \sum_{h=1}^{L} w_h^2 \frac{1-f_h}{n_h} S_{yh}^2 (1-\rho_h^2), \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{yLS}) = \sum_{h=1}^{L} w_h^2 \frac{1-f_h}{n_h} MS_{E,h},$$

where $MS_{E,h}$ is the residual mean square of the regression of y on x in stratum h.

Step 4. As for the separate ratio estimator, the approximation of Step 2 needs every n_h to be large; with small strata the slope errors (and the resulting biases) accumulate over the L strata. ■

Exercise 7.2 (Combined regression estimator with a given slope)

With a slope β fixed in advance (the same for all strata), let $\hat{\mu}_{yLC} = \bar{y}_{\text{st}} + \beta(\mu_x - \bar{x}_{\text{st}})$.
(a) Show that $\hat{\mu}_{yLC}$ is unbiased and find $\text{Var}(\hat{\mu}_{yLC})$.
(b) Find the β minimizing the variance and compare $\hat{\mu}_{yLC}$ with the combined ratio estimator $\hat{\mu}_{yRC}$.

Solution. Write $a_h = w_h^2(1 - f_h)/n_h$ and S_{xyh} for the covariance of x and y in stratum h.

(a)

Step 1. $\hat{\mu}_{yLC} = \sum_h w_h \bar{u}_h + \beta\mu_x$ is the stratified mean of $u_{hi} = y_{hi} - \beta x_{hi}$ plus a constant. Since $\text{E}(\bar{y}_{\text{st}}) = \mu_y$ and $\text{E}(\bar{x}_{\text{st}}) = \mu_x$, $\text{E}(\hat{\mu}_{yLC}) = \mu_y$ exactly.

Step 2. By the stratified variance theorem applied to u, whose stratum variance is $S_{yh}^2 - 2\beta S_{xyh} + \beta^2 S_{xh}^2$,

$$\text{Var}(\hat{\mu}_{yLC}) = Q(\beta) := \sum_h a_h (S_{yh}^2 - 2\beta S_{xyh} + \beta^2 S_{xh}^2) = A - 2\beta B + \beta^2 C,$$

with $A = \sum_h a_h S_{yh}^2$, $B = \sum_h a_h S_{xyh}$, $C = \sum_h a_h S_{xh}^2 > 0$.

(b)

Step 1. Q is minimized at $\beta_{\text{opt}} = B/C = \sum_h a_h S_{xyh} / \sum_h a_h S_{xh}^2$, a weighted average of the within-stratum covariances over variances, and completing the square gives

$$Q(\beta) = Q(\beta_{\text{opt}}) + C(\beta - \beta_{\text{opt}})^2.$$

Step 2. By the classbox on separate versus combined ratio estimators in Ch. 5, $\text{Var}(\hat{\mu}_{yRC}) \approx \sum_h a_h S_{h,C}^2$ with $S_{h,C}^2 = S_{yh}^2 - 2R_C\rho_h S_{xh} S_{yh} + R_C^2 S_{xh}^2$, i.e. $\text{Var}(\hat{\mu}_{yRC}) \approx Q(R_C)$: to first order the combined ratio estimator is the combined regression estimator with slope $R_C = \mu_y/\mu_x$.

Step 3. Therefore

$$\text{Var}(\hat{\mu}_{yRC}) - \text{Var}(\hat{\mu}_{yLC})\big|_{\beta=\beta_{\text{opt}}} \approx C\,(R_C - \beta_{\text{opt}})^2 \;\geq 0,$$

with equality iff $R_C = \beta_{\text{opt}}$, i.e. when the pooled within-stratum regression line passes through the origin. In practice β_{opt} is estimated by the pooled sample slope, which changes the variance only by a term of smaller order. ■

Exercise 7.3 (Exact bias of the regression estimator)

Under SRS, assume the sample x's are never all equal, so b is defined. Show that

$$\text{E}(\hat{\mu}_{yL}) - \mu_y = -\text{Cov}(b, \bar{x}), \qquad |\text{E}(\hat{\mu}_{yL}) - \mu_y| \leq \sqrt{\text{Var}(b)}\,\sqrt{\frac{1-f}{n}}\,S_x,$$

and explain why the bias is negligible relative to the standard error for large n.

Solution. [Supplement]

Step 1. $\text{E}(\bar{y}) = \mu_y$ and $\text{E}(\bar{x} - \mu_x) = 0$ under SRS, so

$$\text{E}(\hat{\mu}_{yL}) - \mu_y = -\text{E}[b(\bar{x} - \mu_x)] = -\{\text{E}[b(\bar{x} - \mu_x)] - \text{E}(b)\,\text{E}(\bar{x} - \mu_x)\} = -\text{Cov}(b, \bar{x}).$$

Step 2. By the Cauchy–Schwarz inequality $|\text{Cov}(b, \bar{x})| \leq \sqrt{\text{Var}(b)\text{Var}(\bar{x})}$, and $\text{Var}(\bar{x}) = \frac{1-f}{n} S_x^2$.

Step 3. Both $\sqrt{\mathrm{Var}(b)}$ and $\sqrt{\mathrm{Var}(\bar{x})}$ are of order $n^{-1/2}$, so the bias is of order $1/n$, while the standard error of $\hat{\mu}_{yL}$ is of order $n^{-1/2}$. This is the exact analogue of the bias bound $|\mathrm{E}(r) - R| \le CV(\bar{x})\sqrt{\mathrm{Var}(r)}$ for the ratio estimator. The bias vanishes when b is uncorrelated with $\bar{x}$, e.g. under the linear model with fixed x's (the model-unbiasedness theorem). ■

Exercise 7.4 (The regression estimator as a calibrated weighting)

Let $S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 > 0$ and $g_i = 1 + \dfrac{n(\mu_x - \bar{x})(x_i - \bar{x})}{S_{xx}}$.

(a) Show that $\hat{\mu}_{yL} = \frac{1}{n}\sum_i g_i y_i$.

(b) Show that $\frac{1}{n}\sum_i g_i = 1$ and $\frac{1}{n}\sum_i g_i x_i = \mu_x$: the weights reproduce the known mean of x exactly.

(c) Show that, among all weights (g_i') satisfying the two equations in (b), the g_i minimize $\sum_i (g_i' - 1)^2$.

Solution. [Supplement]

(a) $b = \sum_i (x_i - \bar{x})y_i / S_{xx}$, so $\bar{y} + b(\mu_x - \bar{x}) = \frac{1}{n}\sum_i y_i\big[1 + n(\mu_x - \bar{x})(x_i - \bar{x})/S_{xx}\big] = \frac{1}{n}\sum_i g_i y_i$.

(b) Since $\sum_i (x_i - \bar{x}) = 0$, $\sum_i g_i = n$. Since $\sum_i (x_i - \bar{x})x_i = S_{xx}$, $\sum_i g_i x_i = n\bar{x} + n(\mu_x - \bar{x}) = n\mu_x$.

(c)

Step 1. Put $h_i = g_i - 1 = \lambda(x_i - \bar{x})$ with $\lambda = n(\mu_x - \bar{x})/S_{xx}$, and let g_i' be any feasible weights, $d_i = g_i' - g_i$. Subtracting the constraints for g' and g, $\sum_i d_i = 0$ and $\sum_i d_i x_i = 0$.

Step 2. Then $\sum_i h_i d_i = \lambda\big(\sum_i d_i x_i - \bar{x}\sum_i d_i\big) = 0$, so

$$\sum_i (g_i' - 1)^2 = \sum_i (h_i + d_i)^2 = \sum_i h_i^2 + \sum_i d_i^2 \ge \sum_i h_i^2,$$

with equality iff $d = 0$.

Thus $\hat{\mu}_{yL}$ is the weighted mean whose weights stay as close as possible to the SRS weights 1 while being calibrated to the known μ_x. If $\bar{x} < \mu_x$, units with large x receive weights above 1, which is the upward correction described at the start of the chapter. ■